Erratum: G4 / G5-bis power sentences — frozen snapshot

docs/osf-prereg-erratum-g4-power-2026-10-23.md at commit 14c3d5f86443 (file SHA-256 c15c202f…1c1371ed)

Erratum: power sentences in locked G4 / G5-bis pre-registrations

Split: 2026-10-28, moved unchanged from branch `docs/osf-prereg-2026-10` (PR #41) so it can be filed without waiting for E1. **Date:** 2026-10-23 **Status:** erratum note. The locked pre-registrations listed below are **not edited**; this file records the corrected arithmetic. It should be filed on OSF next to them by the PI. **Reproduction:** `scripts/prereg_power_2026_10.py`, section 2 (exact noncentral t, equal-variance two-sample design, equal n per arm, 80 % power). Values are Cohen's d; Hedges' g = d × J(2n − 2). The locked runs used Welch tests; the equal-variance noncentral t is the standard planning approximation for them (as the locked text itself states for quinto / sexto) and is exact only under equal variances.

What is wrong

Locked file (line)	Locked sentence	Correct value
`docs/osf-prereg-g4-quinto-pilot.md:77`	N = 30 per arm at α = 0.0167 detects |g| ≥ 0.74 (Welch two-sided)	two-sided α = 0.0167: **d = 0.857** (g ≈ 0.846); one-sided: d = 0.782 (g ≈ 0.772)
`docs/osf-prereg-g4-sexto-pilot.md:110`	same sentence	same correction
`docs/osf-prereg-g4-septimo-pilot.md:80`	N = 30 at α = 0.05 detects |g| ≥ 0.74 (two-sided)	correct as a d value (d = 0.736); g ≈ 0.726
`docs/osf-prereg-g4-octavo-pilot.md:80`	same as septimo	as septimo
`docs/osf-prereg-g5-bis-richer-esnn.md:106-107,141`	N = 10, two-sided α = 0.0125 detects g ≈ 1.27	two-sided: **d = 1.638** (g ≈ 1.568); one-sided (the test G5-bis actually runs): d = 1.484 (g ≈ 1.421)
`docs/osf-prereg-g5-bis-richer-esnn.md:150`	N = 30 follow-up floor g ≈ 0.74	at the G5-bis α = 0.0125: one-sided d = 0.814 (g ≈ 0.803), two-sided d = 0.886 (g ≈ 0.875)

The 0.74 figure is the α = 0.05 two-sided value. It was carried into the quinto / sexto pre-registrations after their α was lowered to 0.0167 by Bonferroni, without being recomputed. The 1.27 figure in G5-bis does not match any of its stated settings.

Consequences

- No verdict changes. Every affected hypothesis was decided by its pre-registered test and threshold, not by the power sentence.
- The quoted detection floors were optimistic. At N = 30 and α = 0.0167 an effect had to be about d = 0.86 (two-sided) to be detected with 80 % probability, not 0.74. Readings that cite “N = 30 detects |g| ≥ 0.74” as evidence that smaller effects are absent should cite the corrected floor. In particular, a non-rejection in those runs is absence of evidence below d ≈ 0.86, not evidence of absence.
- New drafts (G4-nono, DR-3 probe, E1 controls, 2026-10-23) use the corrected arithmetic.